

Active Directory Enumeration (Without Exploitation)

Why AD is the prey

Most enterprise Windows environments = AD. AD is where attacker value lies: users, groups, trusts, Kerberos tickets, group policies. Adversaries take days here – you learn it in a week.

Unauthenticated enumeration (no creds)

```
# nmap LDAP scripts
nmap -p 389,636 --script ldap-search,ldap-rootdse 10.0.0.30

# null session queries (legacy)
ldapsearch -x -h 10.0.0.30 -b "dc=domain,dc=local" "(objectClass=user)" sAMAccountName

# dnsrecon + nslookup against DC
```

Authenticated enumeration (with creds)

```
# BloodHound (graph AD paths to 'Domain Admin')
bloodhound-python -u user -p 'pass' -d corp.local -ns 10.0.0.30 -c All
# then visualize the AD graph

# PowerView (PowerShell, but can run from Kali via wine/Invoke)
# but for Kali: ldapsearch with authenticated bind

# kerberos user enum (no creds) -> AS-REP Roast check
kerbrute userenum --dc corp.local -d corp.local users.txt
```

GPP / CPassword / stored creds

SYSVOL may contain Group Policy Preferences (GPP) with AES-encrypted passwords

AES key is public (MS14-025)

Find with: `find \\dc\sysvol -name Groups.xml -type f`

Decrypt with: `gpp-decrypt <CPassword>`

Trusts & lateral paths

```
ldapsearch ... "(objectClass=trustedDomain)" # trust relationships
```

A forest trust or domain trust is a privileged pathway – map it.

Opsec of AD enumeration

Loud on the DC logs (Event IDs 4624, 4768, 4771...)

Use a dedicated test lab DC (hyperv-inception or ESXi)

Never run AD attacks on a real employer's prod DC without written RoE

Professional AD enumeration = documenting the attack graph before you attack. Maps save time and provide evidence.

[← From Vulnerability Scan to E...](#)

Module complete ✓